

Architecture Overview

Microsoft Copilot · Azure AI · Fabric · Governance

Flow diagrams below are textual renderings of the Mermaid sources in the repository architecture/ folder. They are sanitized reference architectures for discussion with technical stakeholders.

1. Enterprise Copilot

Sanitized Reference Architecture

```
Employee → Entra ID → Copilot Studio
→ Orchestration / grounding instructions
→ Query embedding → Vector index → ACL / metadata filter → re-rank
→ Azure OpenAI (grounded generation) → Citations → Employee
Knowledge sources: SharePoint · Approved documents · Dataverse
```

Purpose: identity-aware internal Q&A; with hallucination controls and verifiable sources.

2. Enterprise RAG

Sanitized Reference Architecture

```
Documents → Preprocess / classify → Chunk + metadata → Embeddings → Vector index
Query → AuthZ filter → Embed → Semantic/hybrid retrieve → Metadata filter
→ Prompt/context assembly → Azure OpenAI → Grounded answer + citations
→ Evaluate groundedness · relevance · completeness · latency
```

Purpose: operable retrieval pipeline with release gates suitable for financial-services style corpora.

3. Agentic Workflow

Sanitized Reference Architecture

```
User → Copilot Studio → Orchestrator (tool calling)
Tools: Knowledge (RAG) · Data (Dataverse/Databricks API) · Action (REST/MCP)
· Power Automate · Specialist agent
High-risk write? → Human approval → Execute · else Execute
→ Unified response + audit trail
```

Purpose: controlled multi-tool automation with narrowly scoped permissions.

4. Fabric + Copilot

Sanitized Reference Architecture

Sources → Fabric pipelines / Dataflows → Lakehouse
Bronze → Silver → Gold → Semantic model
→ Power BI and Copilot / downstream AI
Governance: Purview · Workspace RBAC · RLS

Purpose: AI consumes curated business data and shared metric definitions—not raw landing tables.

5. Security & Governance

Sanitized Reference Architecture

User → Entra ID → Conditional Access → Copilot / AI app
→ Policy (RBAC · tool allow-list · DLP)
→ Read tools (least privilege) / Write tools (separate identity + approval)
→ Enterprise systems · Purview · Audit logging
Environments: Dev · Test · UAT · Prod

Principle: An AI agent should never provide access to information or operations that the authenticated user would not normally be authorized to access.

Technical Depth Checklist

Concern	Typical control
Grounding	Context-only prompts; citation required
Retrieval AuthZ	ACL/group filter before generation
Tool calling	Schema validation; allow-listed tools
Secrets	Key Vault / pipeline secrets; managed identity
Data for AI	Fabric Gold / semantic model preferred
Promotion	Managed solutions + evaluation evidence

Component Responsibilities

Component	Responsibility
Copilot Studio	Channel UX, topics, generative orchestration
Azure OpenAI	Embeddings, grounded completion, tool selection
Vector index	Semantic retrieval with metadata payload
Power Automate	Workflows, approvals, Platform connectors
Dataverse	Operational entities and configuration
SharePoint	Document knowledge corpus
Fabric Lakehouse	Medallion analytics foundation
Semantic model	Measures, relationships, RLS for BI/AI
Entra ID	Authentication, groups, conditional access
Purview	Classification, catalog, lineage signals
Key Vault	Secret/certificate material when required
CI/CD	Solution validation and environment promotion

Integration Notes

- Custom connectors expose OpenAPI-defined REST surfaces to Copilot and flows.
- MCP-based integrations may standardize tool access alongside connectors.
- OAuth 2.0 delegated calls are preferred when the action must run as the user.
- Databricks and SAP are reached through governed APIs—not credentials in prompts.
- Evaluation datasets gate prompt/index changes the same way tests gate code.

Repository Diagram Sources

Maintainable Mermaid diagrams: [architecture/enterprise-copilot-architecture.md](#), [rag-architecture.md](#), [agentic-workflow.md](#), [fabric-ai-architecture.md](#), [security-governance.md](#). SVG exports for slides/PDF packaging: [assets/diagrams/](#).

End-to-End Sequence (Knowledge + Action)

1. Authenticate the user with Entra ID and establish session claims.
2. Classify intent: knowledge, data lookup, workflow, or mixed.
3. For knowledge: embed query, retrieve with ACL/metadata filters, assemble context, generate grounded answer with citations.
4. For actions: validate tool schema, authorize tool and resource, execute via connector/API/MCP or Power Automate.
5. For high-risk writes: pause for human approval; deny by default on failure.
6. Log correlation ID, tools used, and outcome; return a user-safe response.
7. Feed production telemetry and evaluation samples back into the ALM quality gate.

What Is Intentionally Not Shown

- Client network diagrams, private endpoints, or tenant identifiers
- Proprietary prompt libraries and production index schemas
- Real document corpora, customer records, or internal SharePoint URLs
- Credentials, certificates, connection strings, or pipeline secrets

Confidentiality: Client-specific implementations, source code, datasets, credentials, internal URLs, and proprietary architecture details are intentionally excluded.